

Computational Reproducibility and Data Usage Guide for Transition-State Damping as a Mechanistic Descriptor of Rate Turnover in Chemical Reactivity

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Overview

This document describes the datasets, computational tools, and analysis procedures required to reproduce the calculations presented in the manuscript:

Transition-State Damping as a Mechanistic Descriptor of Rate Turnover in Chemical Reactivity.

The repository contains scripts and datasets required to reproduce:

- the damping ratio χ calculations
- transition-state dynamical regime classifications
- cross-domain statistical clustering analysis (Test 8)
- catalytic dataset construction used in the manuscript figures

All tools are implemented in Python and operate directly on the included CSV datasets. No proprietary software is required.

1 Core χ Computation Methods

The central quantity used throughout the manuscript is the dimensionless damping ratio

$$\chi = \frac{\Gamma_{\text{eff}}}{2\Omega_{\text{TS}}} \quad (1)$$

All χ calculations are implemented in:

`stability_methods.py`

This module provides three estimator routes corresponding to experimental or computational observables.

1.1 Route A: Spectroscopic Estimator

Function:

`chi_route_a(delta_nu_L_cm1, nu_cm1)`

The estimator is

$$\chi = \frac{\Delta\tilde{\nu}_L}{2\tilde{\nu}} \quad (2)$$

This route is used when vibrational damping is measured experimentally using IR or Raman spectroscopy.

1.2 Route B: Solvent Relaxation Estimator

Function:

```
chi_route_b(tau_L_ps, lambda_ev, delta_q_angstrom, m_eff_amu)
```

The coordinate frequency is

$$\Omega = \sqrt{\frac{2\lambda}{m_{\text{eff}}\Delta Q^2}} \quad (3)$$

and the damping ratio becomes

$$\chi = \frac{1}{2\tau_L\Omega} \quad (4)$$

This route applies to solvent-controlled electron transfer or PCET systems.

1.3 Route C: Electronic Friction Estimator

Function:

```
chi_route_c(gamma_cat_ev, nu_cm1, gamma_int_ev)
```

$$\chi = \frac{\Gamma_{\text{cat}} + \Gamma_{\text{int}}}{2hc\tilde{\nu}} \quad (5)$$

This estimator is used primarily for heterogeneous catalysis where friction tensors are obtained from density functional theory calculations.

2 Transition-State χ Evaluation

Input dataset:

```
ts_chi_input.csv
```

Processing script:

```
ts_chi_tools.py
```

Output dataset:

```
chi_ts_results.csv
```

The script reads the transition-state dataset and calculates χ values, then classifies each system into dynamical regimes.

χ Range	Dynamical Regime
$\chi < 0.8$	Underdamped
$0.8 \leq \chi \leq 1.2$	Near-critical
$\chi > 1.2$	Overdamped

3 Cross-Domain Statistical Test (Test 8)

Dataset:

```
test8_cross_domain.csv
```

Script:

```
Test8_pValueCal.py
```

The statistical test evaluates whether optimal systems across independent chemical domains cluster near $\chi \approx 1$ by chance.

The probability window used in the manuscript is

$$0.8 \leq \chi \leq 1.2 \quad (6)$$

assuming a log-uniform distribution across

$$10^{-4} \leq \chi \leq 10^1. \quad (7)$$

For $N = 3$ independent systems the calculation produces

- Single-system probability $\approx 3.52\%$
- Combined probability $\approx 4.37 \times 10^{-5}$

4 Catalysis Dataset Construction

Script:

```
oc20_n2_catalog.py
```

Output dataset:

```
oc20_n2_catalog.csv
```

This script extracts N₂ adsorption systems from the Open Catalyst Project (OC20) metadata and compiles the catalytic screening dataset used in the manuscript.

5 Periodic Table χ Proxy Dataset

Dataset:

```
periodic_table_chi_proxy.csv
```

This dataset contains electron-phonon coupling and Debye temperature values used to compute the lattice damping proxy

$$\chi_{\text{lattice}} = \frac{\pi \lambda_{ep} T}{2\Theta_D}. \quad (8)$$

6 Minimal Reproduction Workflow

Compute transition-state χ values

```
python ts_chi_tools.py ts_chi_input.csv
```

Run the cross-domain clustering test

```
python Test8_pValueCal.py
```

Optionally regenerate the catalytic dataset

```
python oc20_n2_catalog.py
```

Data Availability

All datasets, scripts, and templates required to reproduce the calculations presented in the manuscript are included in the repository. No proprietary data or external software dependencies are required.